

Cross-lags, heterogeneity, and life-history trade-offs



Today is another fun paradox day!

Today we're going to follow 100 female gators around for 7 years



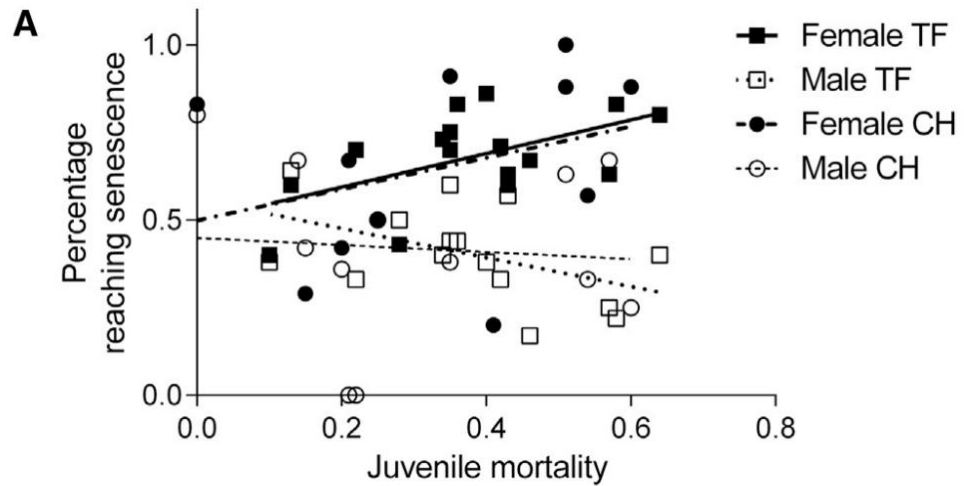
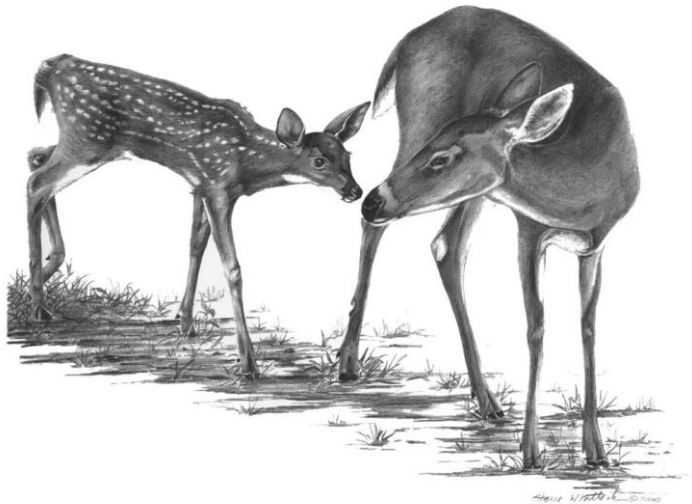
We'll mark individuals, capture them prior to each breeding season to record morphometric data, and visit their nests to measure fecundity.

Female alligators provide parental care

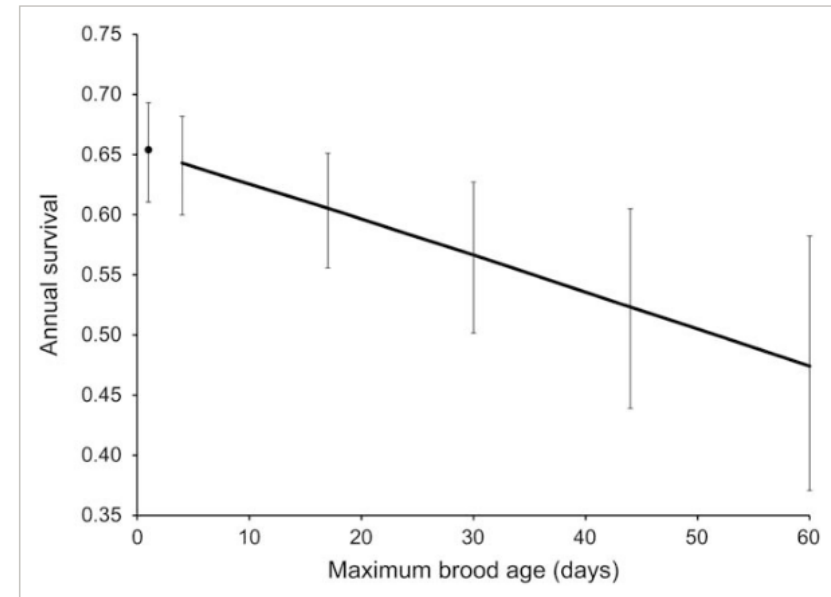


We'll imagine there's a cost to that (i.e., trade-off)

Life-history trade-offs

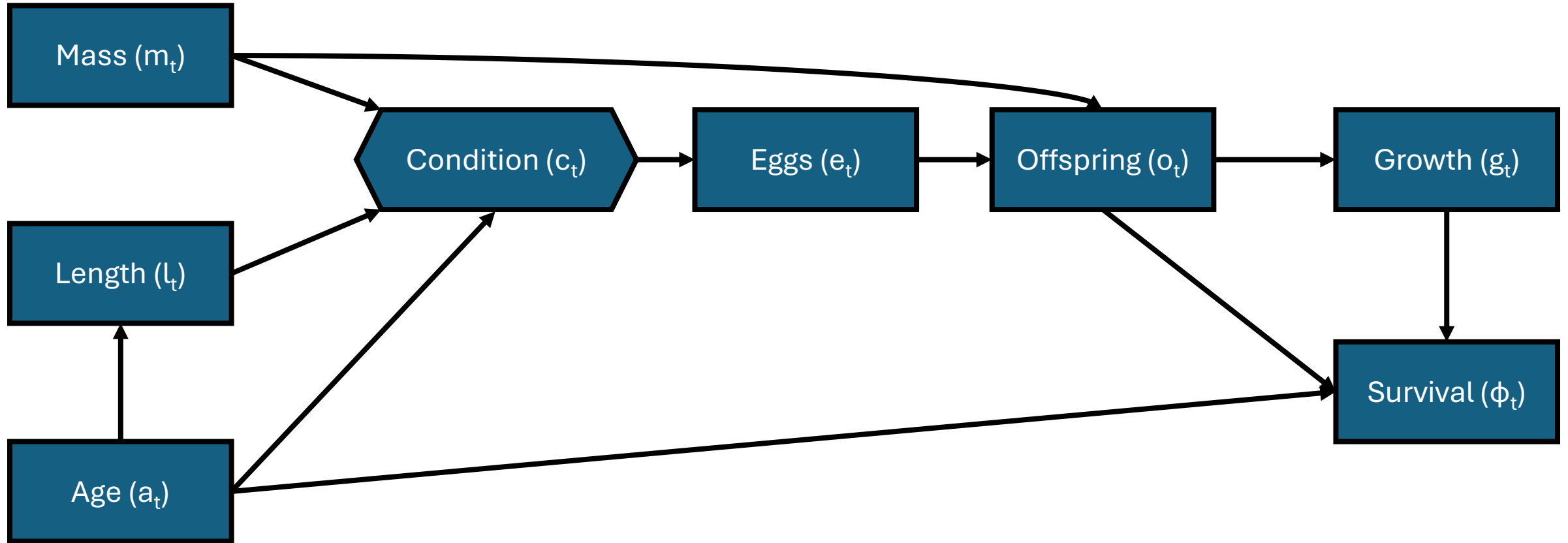


Garratt et al. (2015) Current Biology



Arnold and Howerter (2012) Wildlife Society Bulletin

We could imagine a wildly complicated model here!?



Bühler, Riecke, et al. **(2024)** *Royal Society Open Science*
Layton-Matthews et al. **(2019)** *Global Change Biology*

But today we'll measure just two traits

- Body condition (x ; this will be centered)
- Clutch size (y ; this will be measured in eggs)

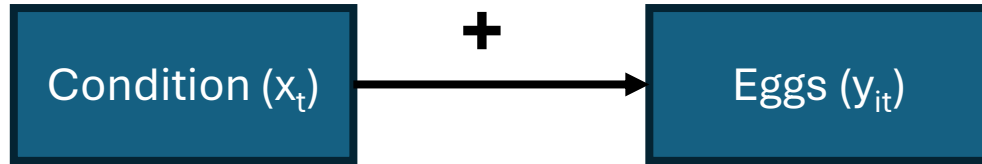
Body condition (x; centered, i.e., $\mu = 0$)



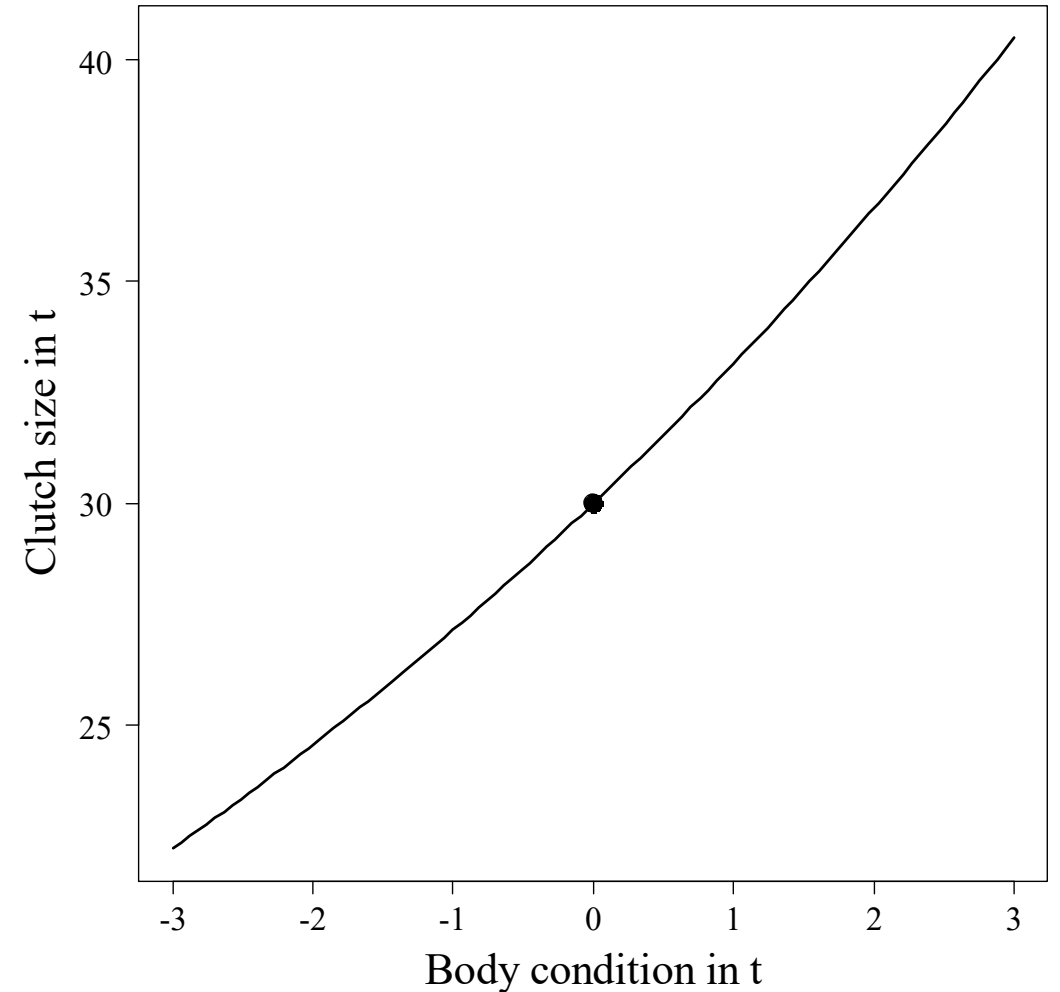
Clutch size (y; number of eggs)



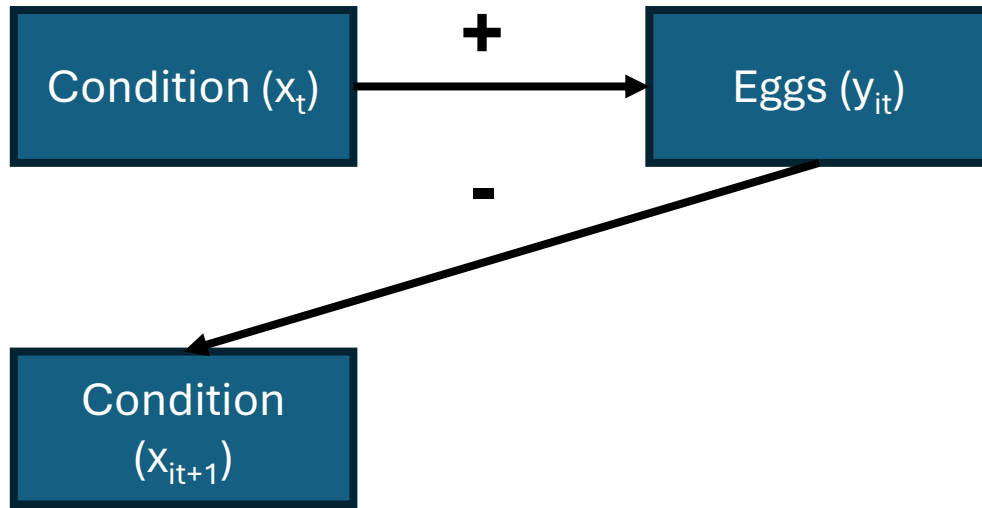
Hypothesis 1: Alligators in better condition will lay more eggs



$$y_{it} \sim \text{Poisson}(e^{\alpha_{i1} + \beta_1 x_{it}})$$

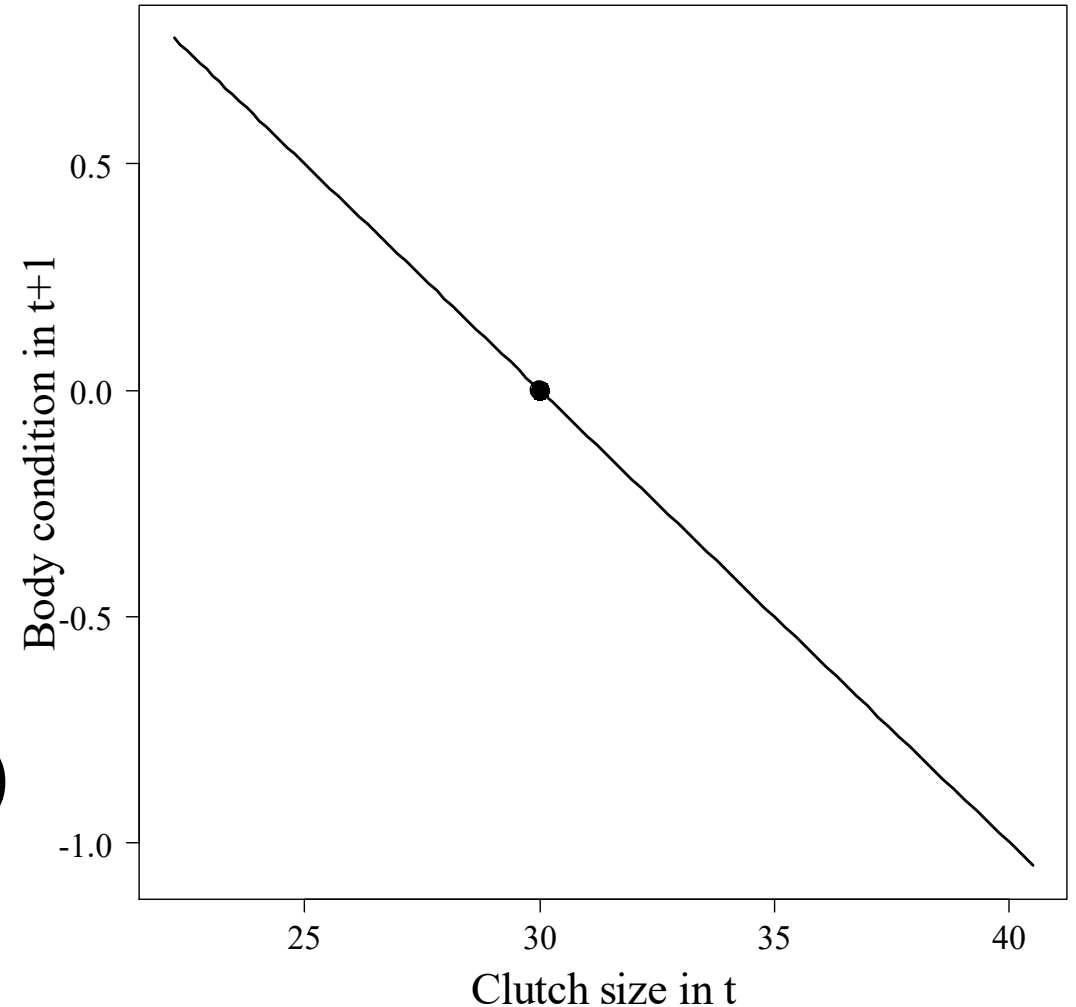


Hypothesis 2: Alligators that lay more eggs will 'lose' condition

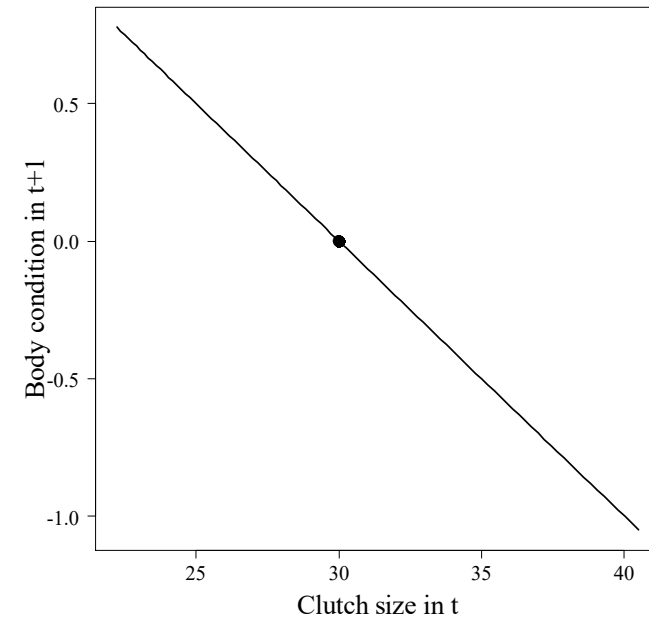
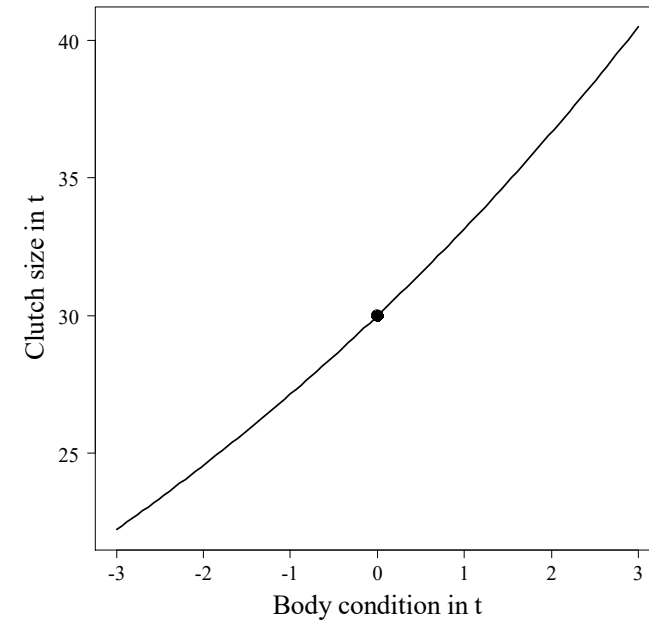
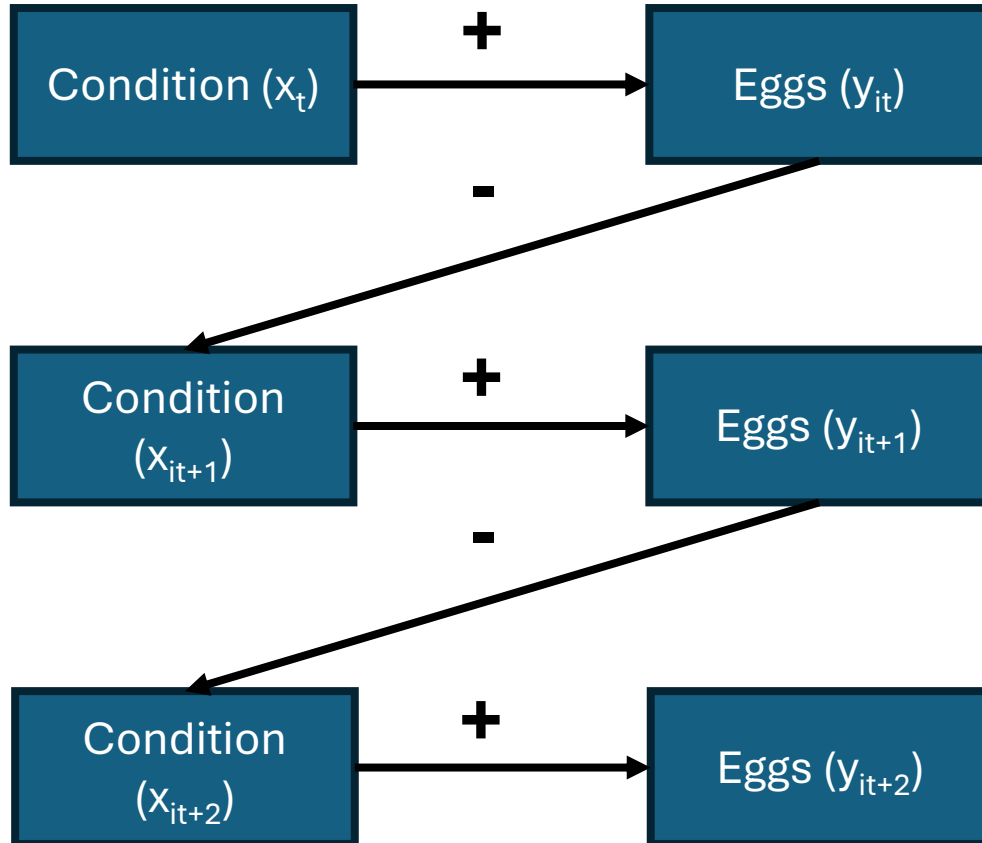


$$E(x_{it+1}) = \alpha_{i2} + \beta_2 \times (y_{it} - e^{\alpha_{i1}})$$

$$x_{it+1} \sim \text{Normal}(E(x_{it+1}), \sigma_x^2)$$

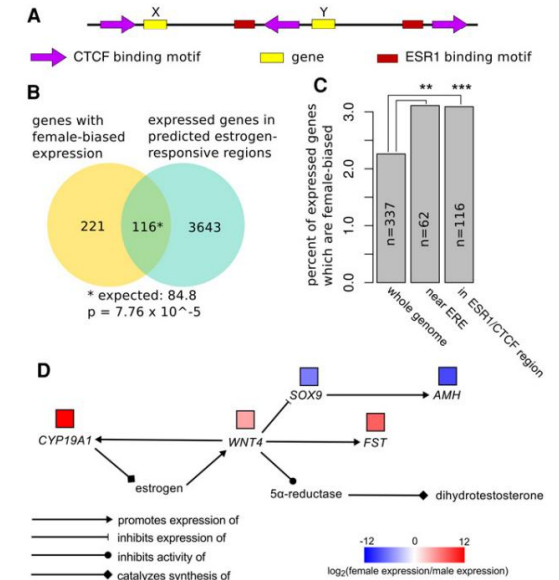
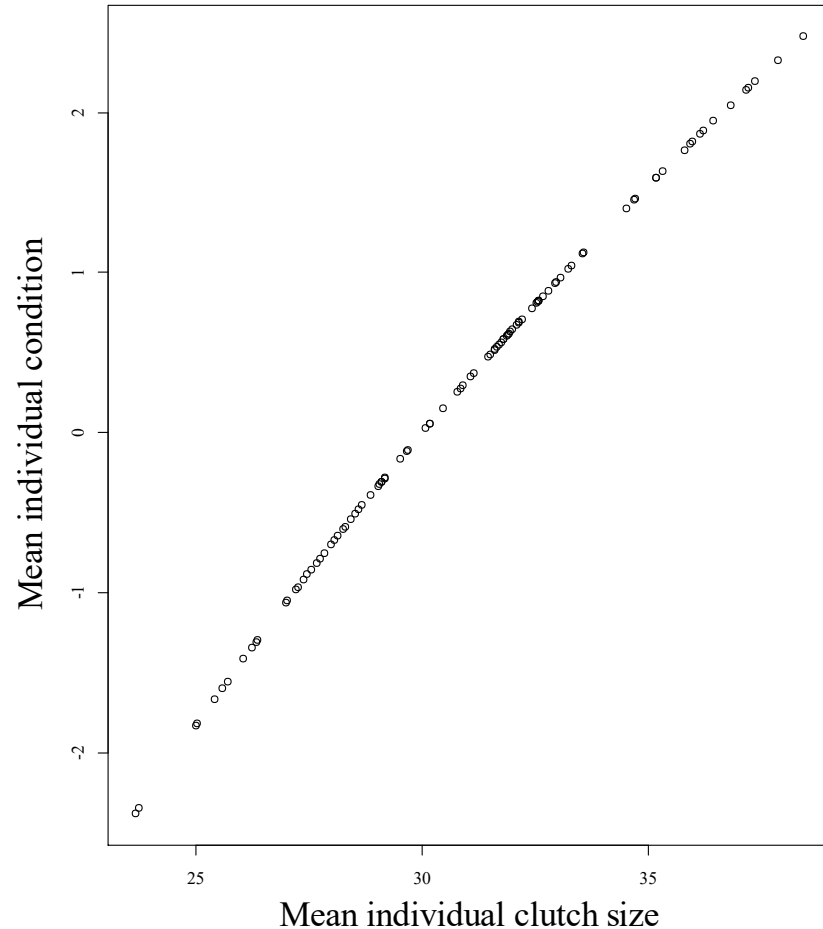


And so on and so forth...



There's one 'wrench' in our plans

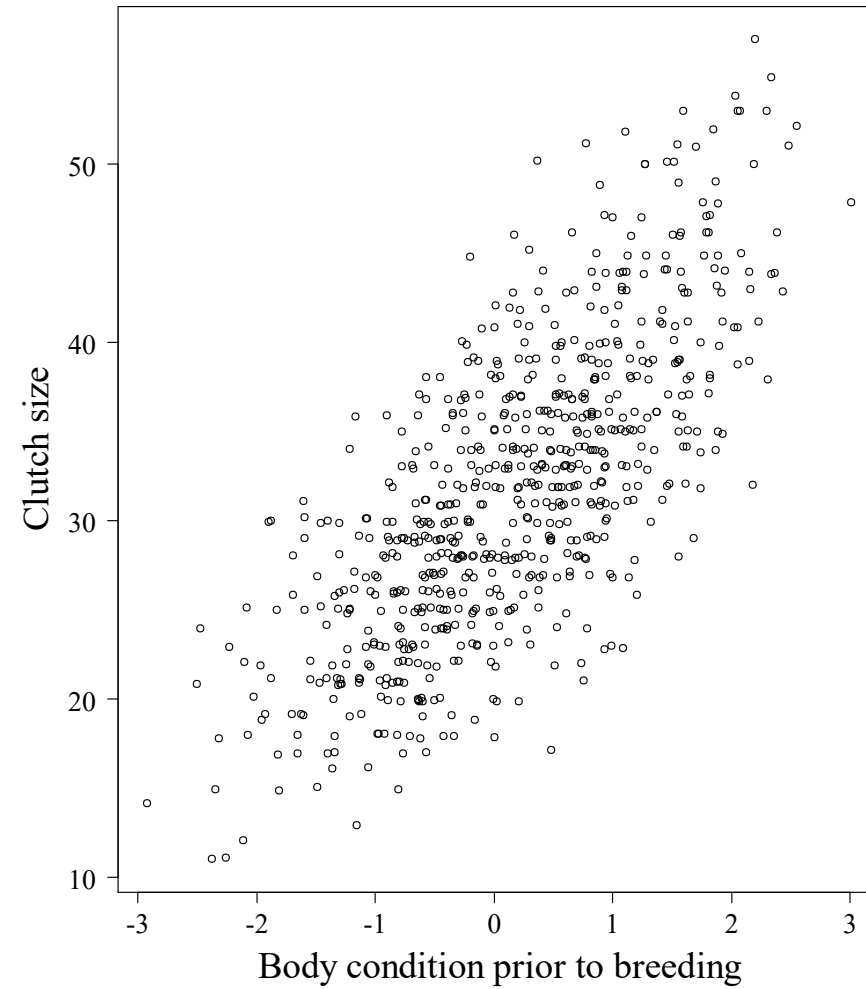
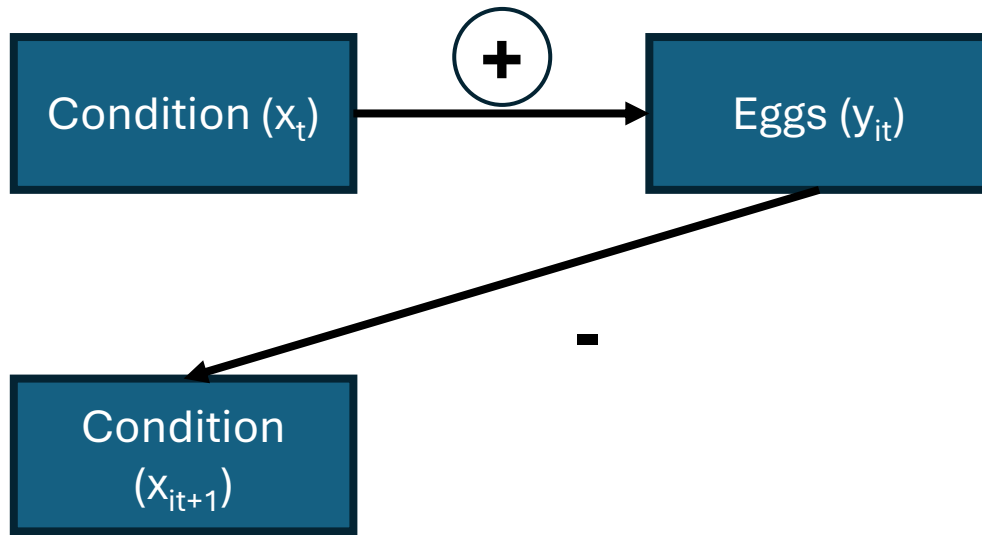
The wrench: These alligators are different (i.e., heterogeneous)...



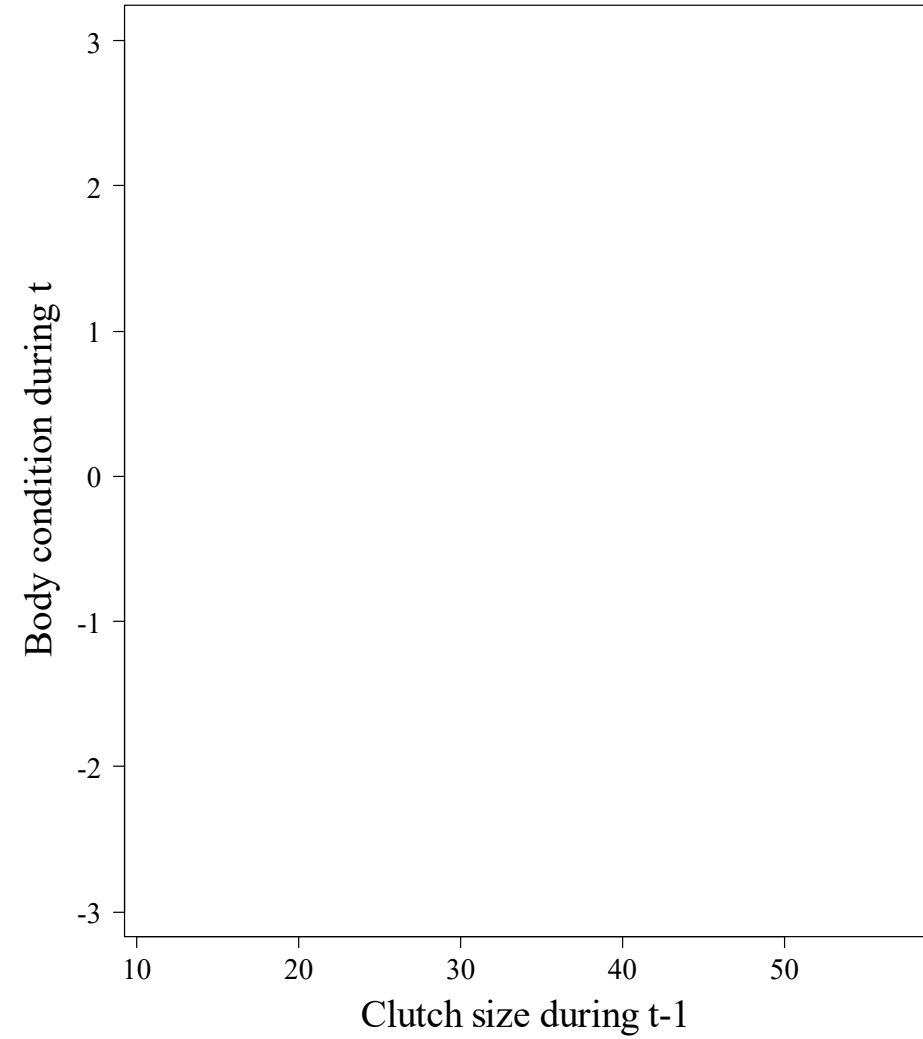
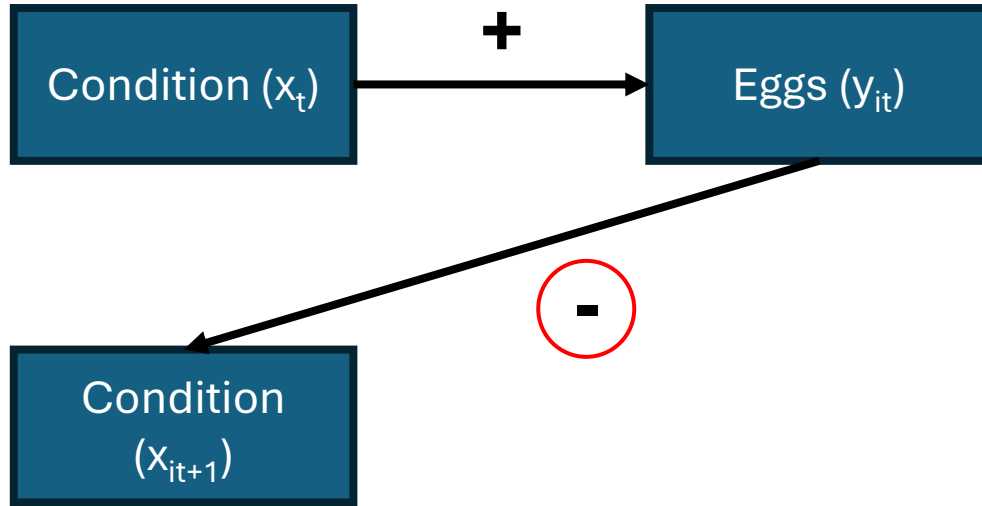
This could be due to unmeasured individual differences or territory quality

So, what do we see at the population level?

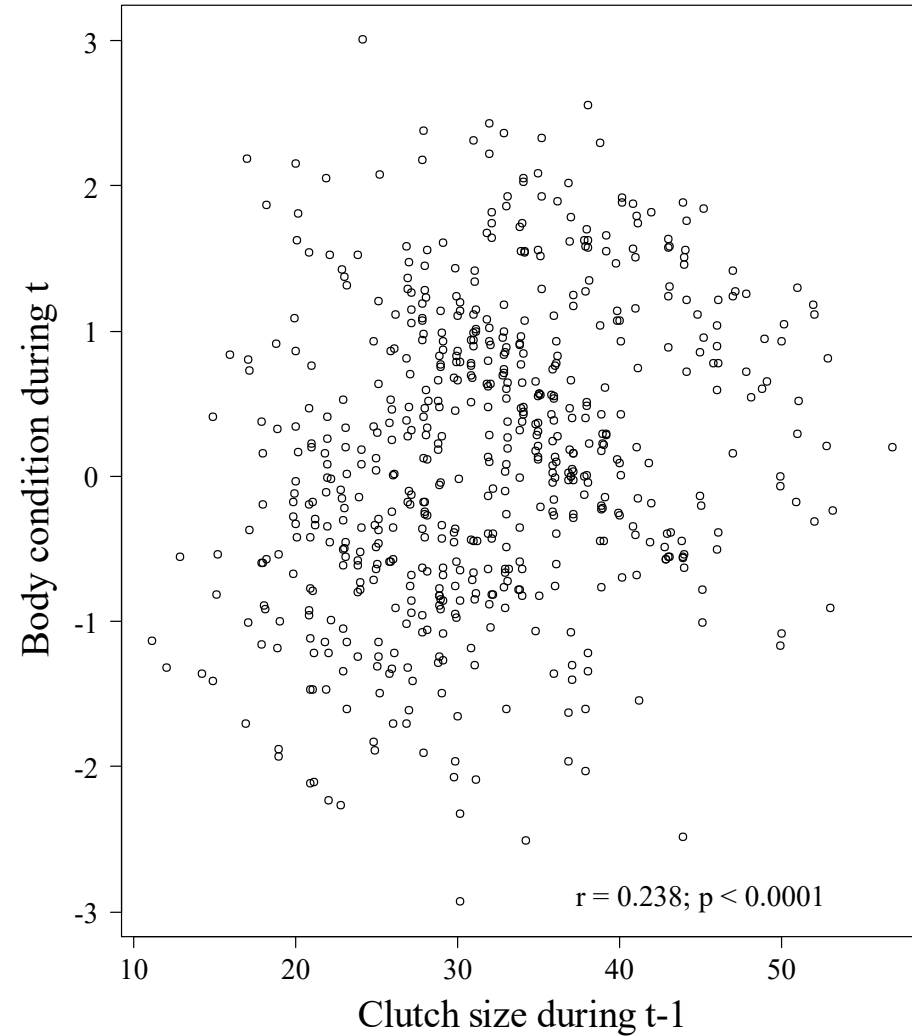
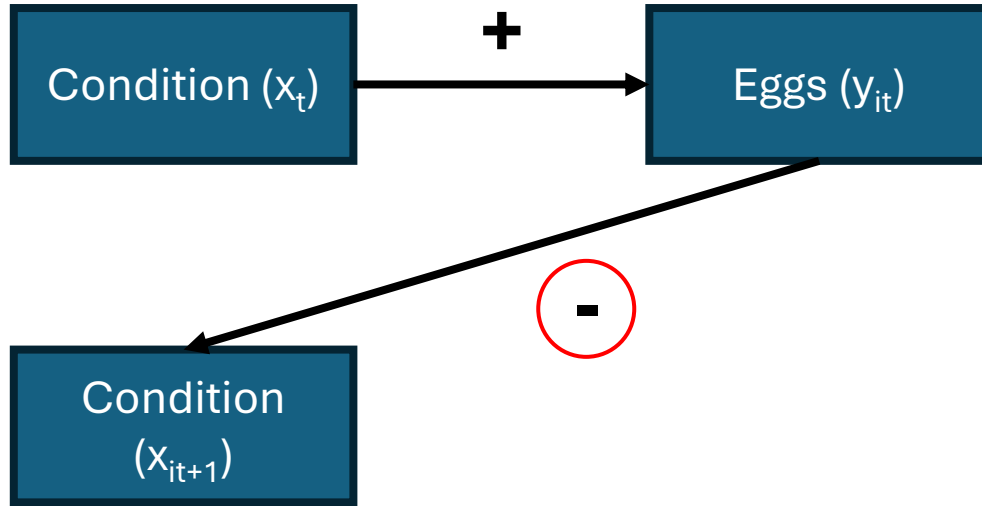
Better condition leads to more eggs in a clutch



Laying more eggs leads to...

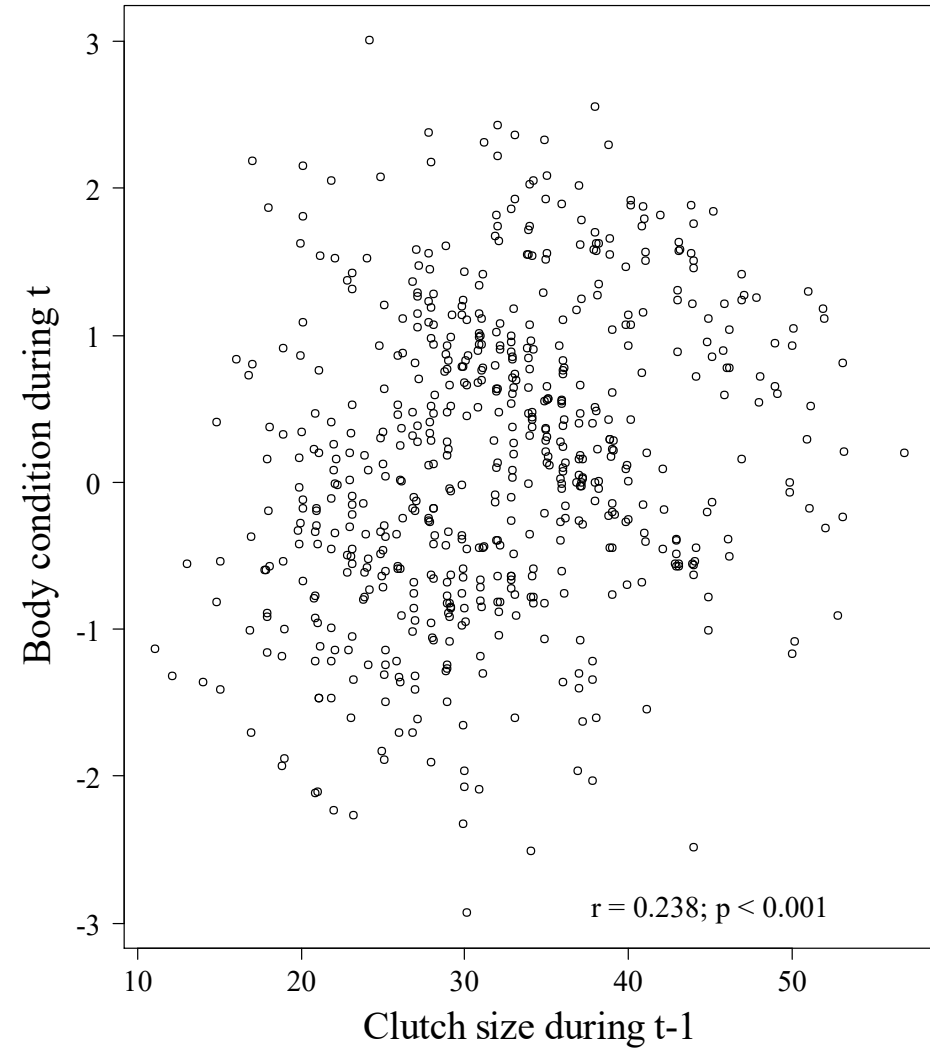
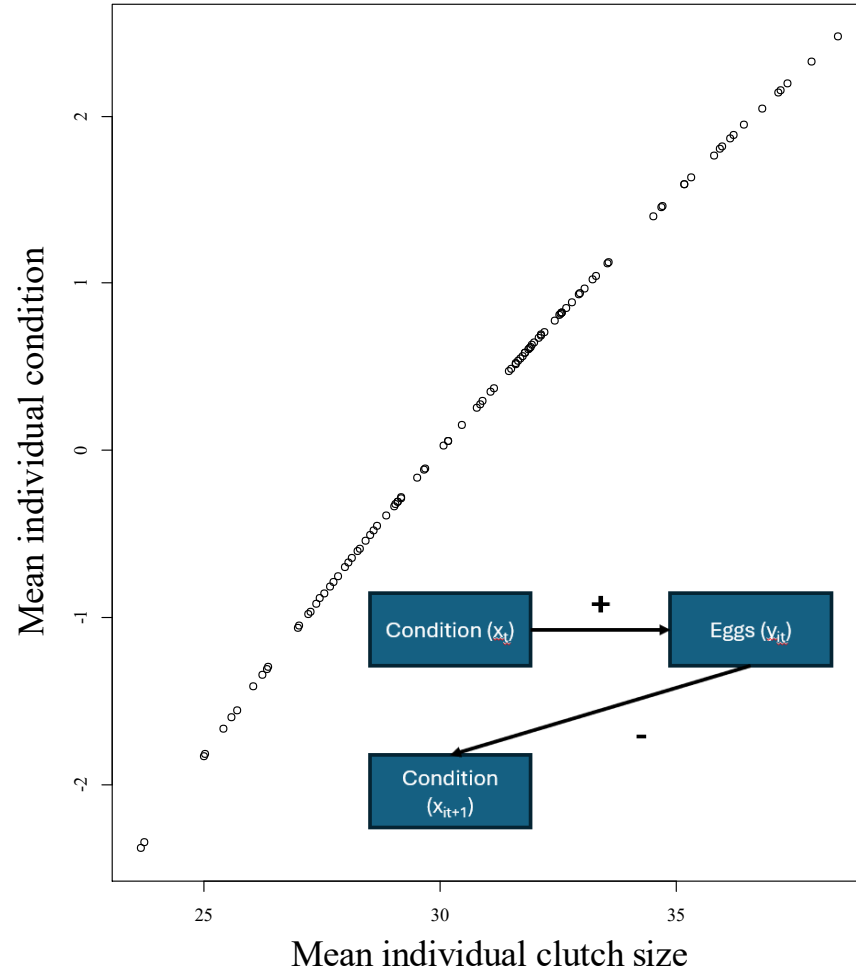


Laying more eggs leads to... better condition next year?!



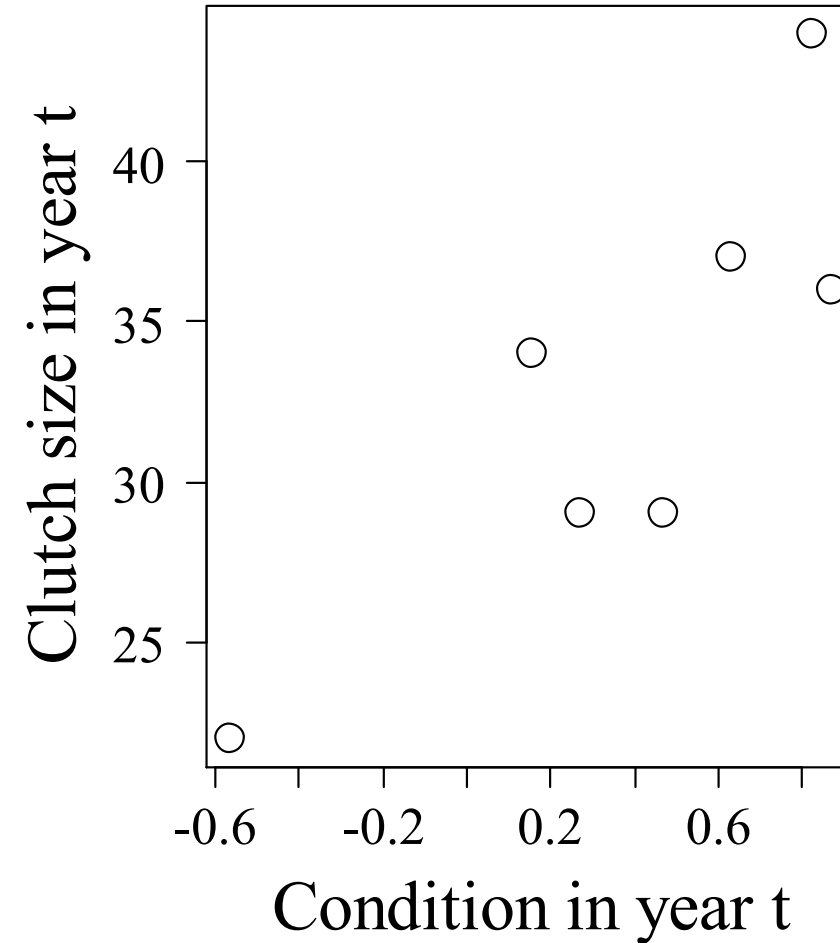
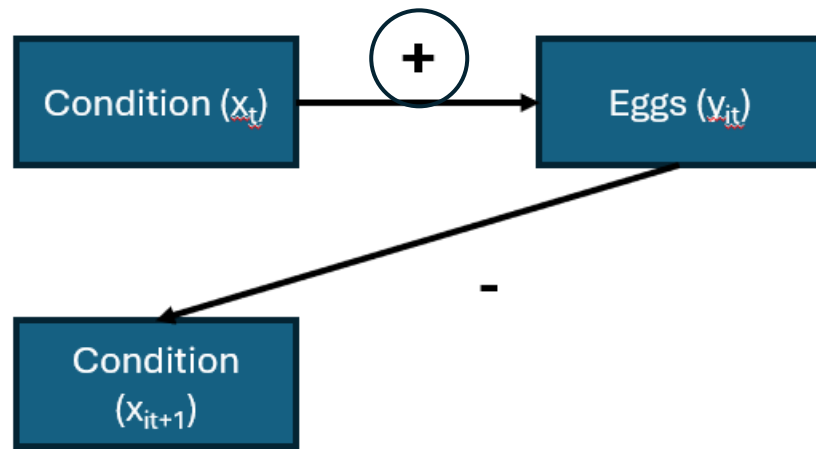
What's happening here?

What's happening here?!



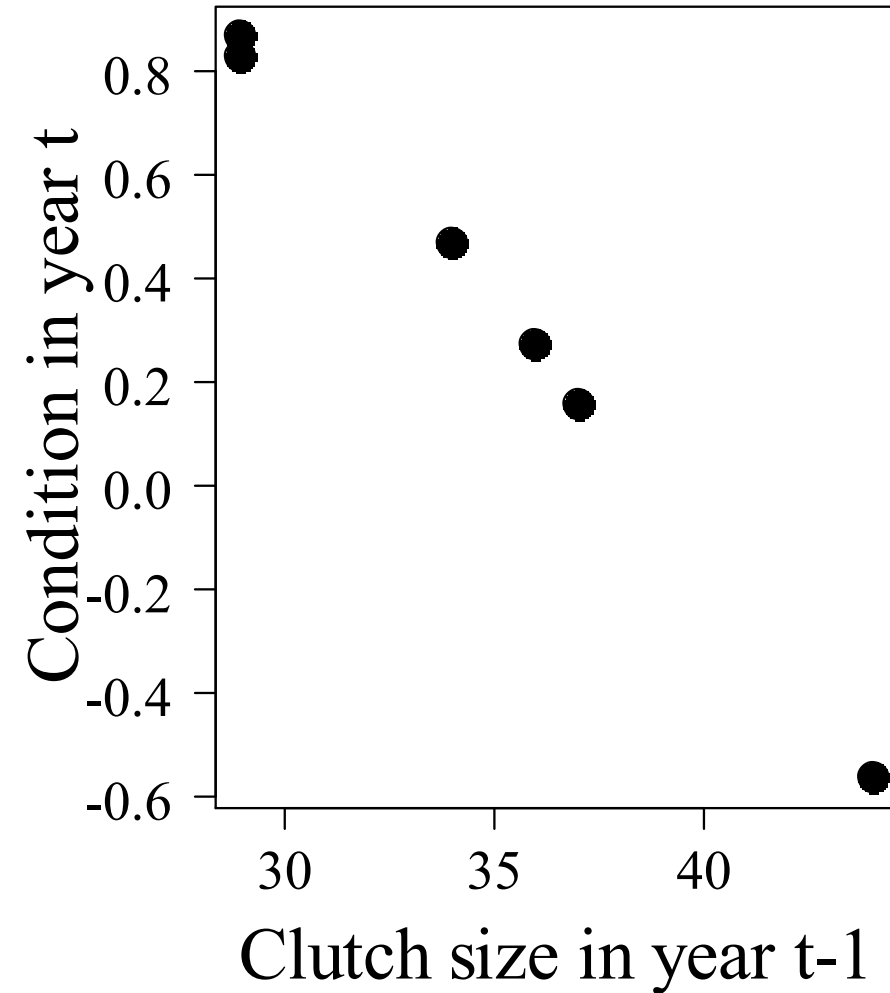
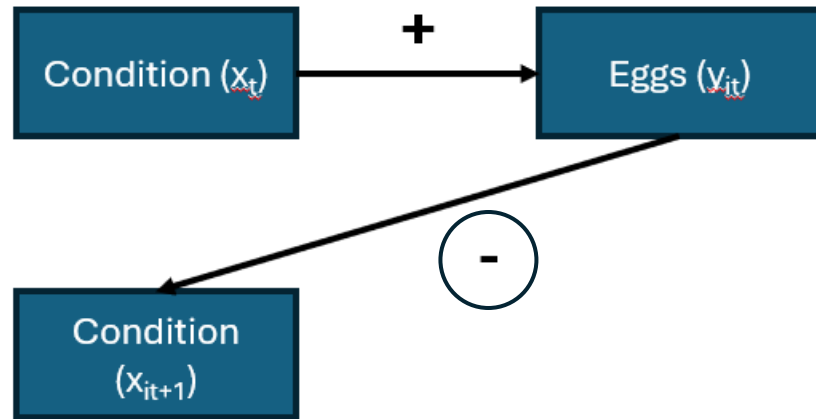
Talk to your neighbor(s), look for patterns in the right-hand figure

What's happening here?! Let's look at an individual



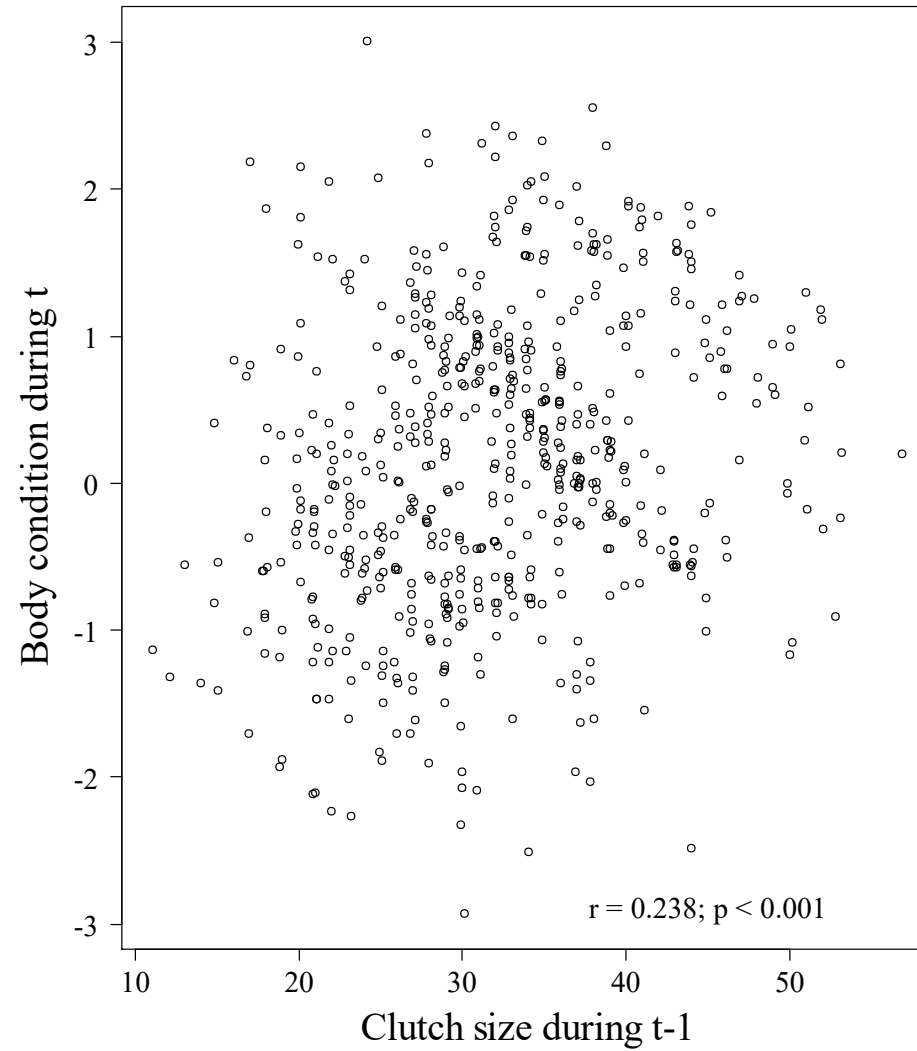
Still a positive effect of condition on clutch size

What's happening here?! Let's look at an individual

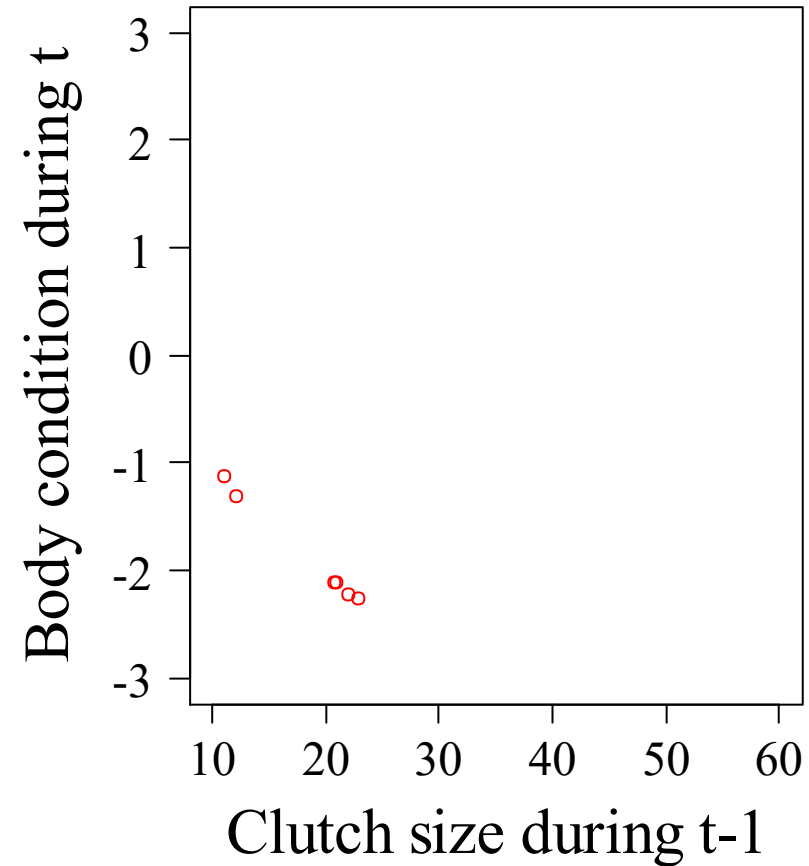
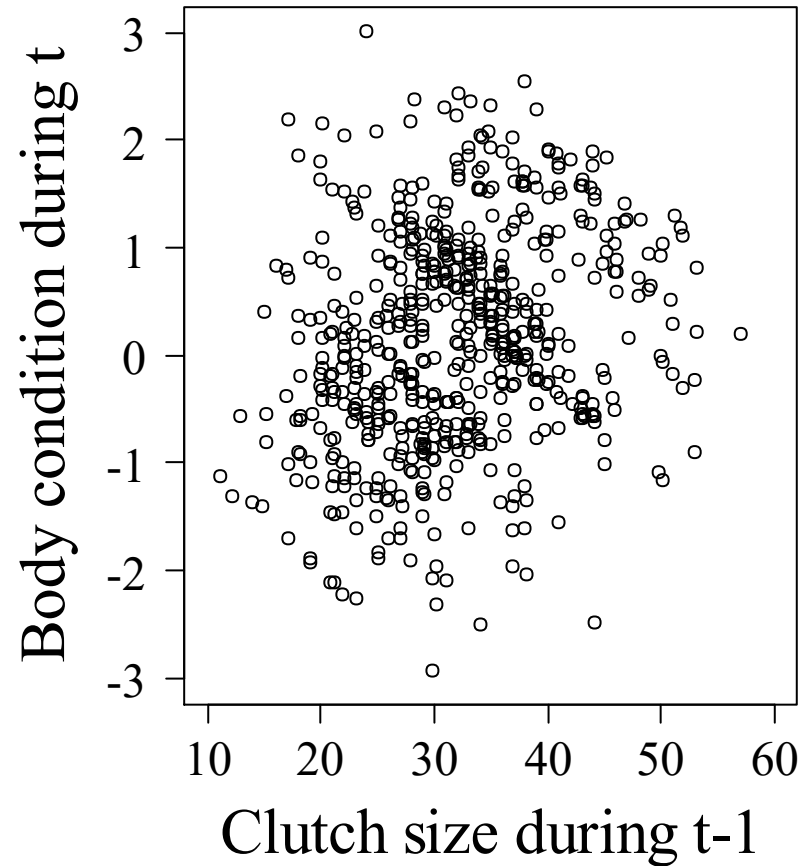


Trade-offs are still happening (cost of reproductive investment)

What's happening here?!

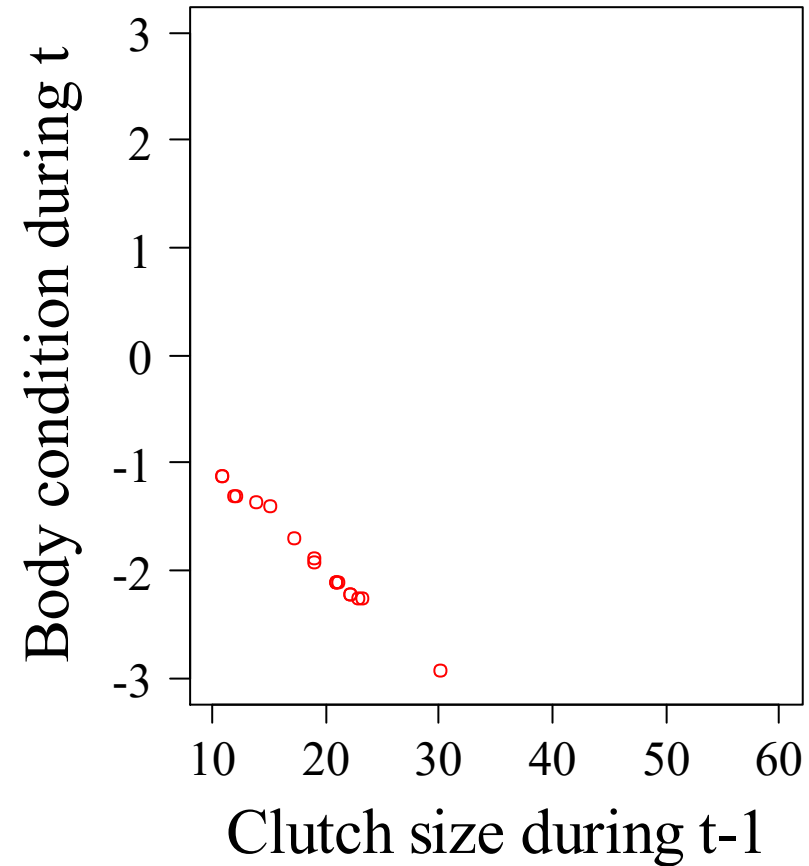
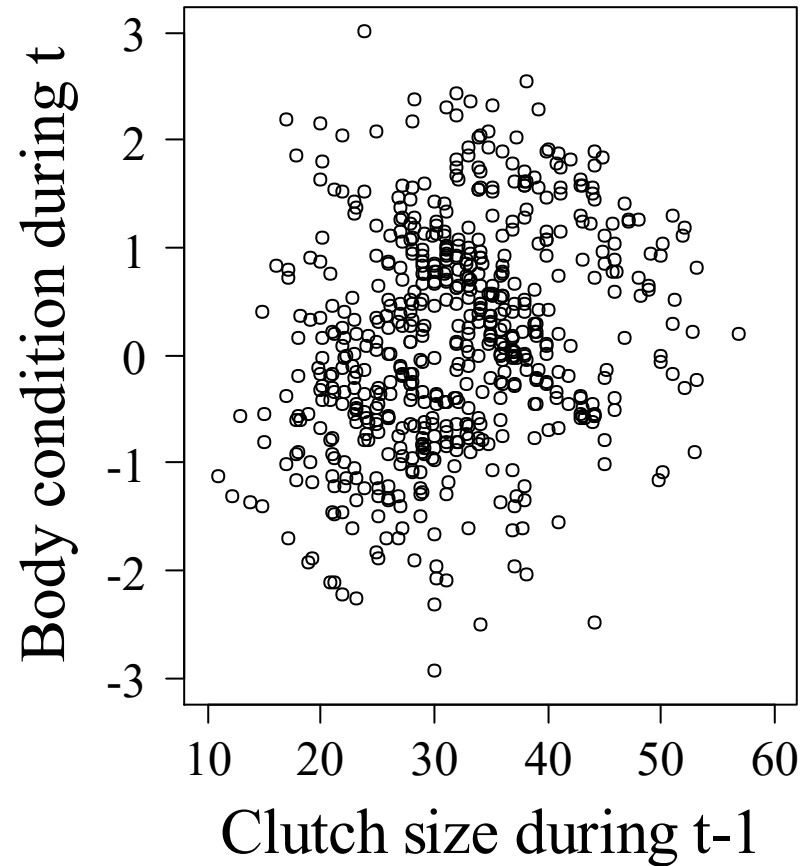


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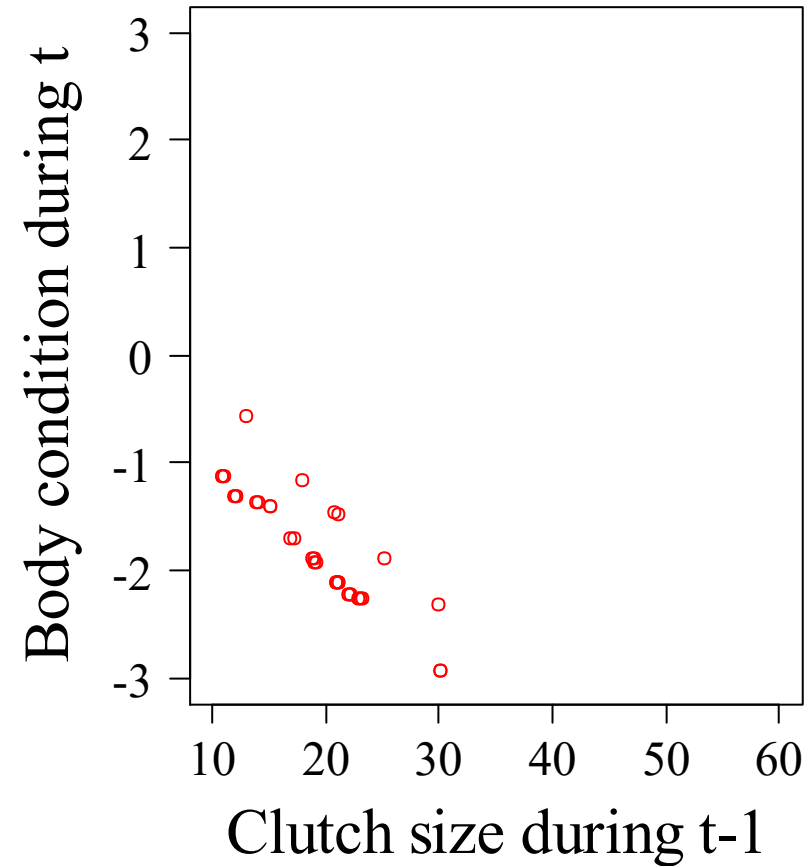
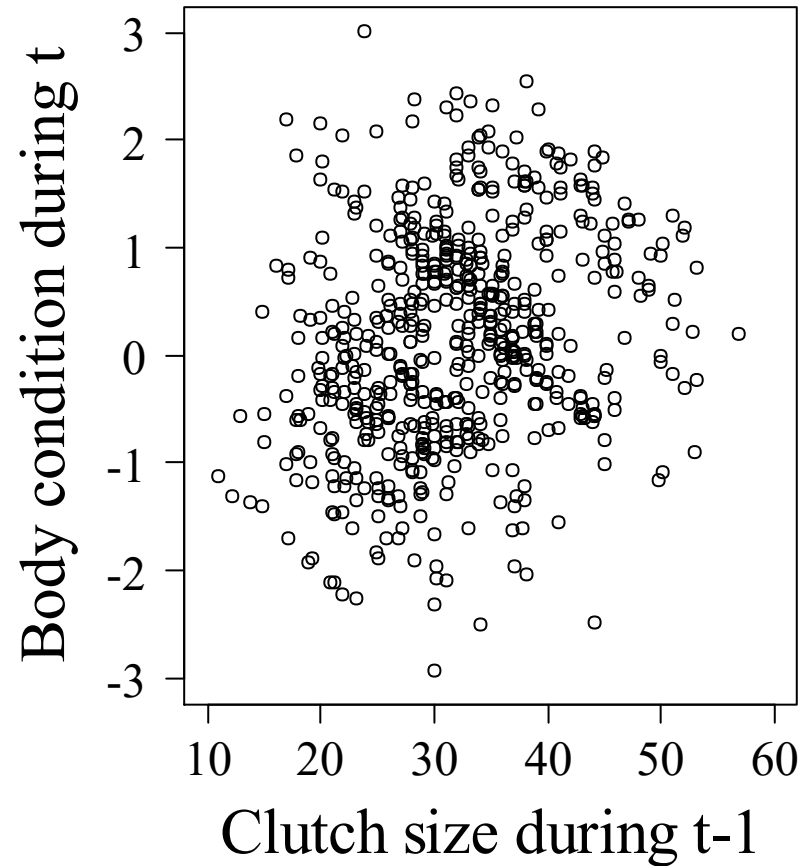
Here is the 'worst' gator

What's happening here?!



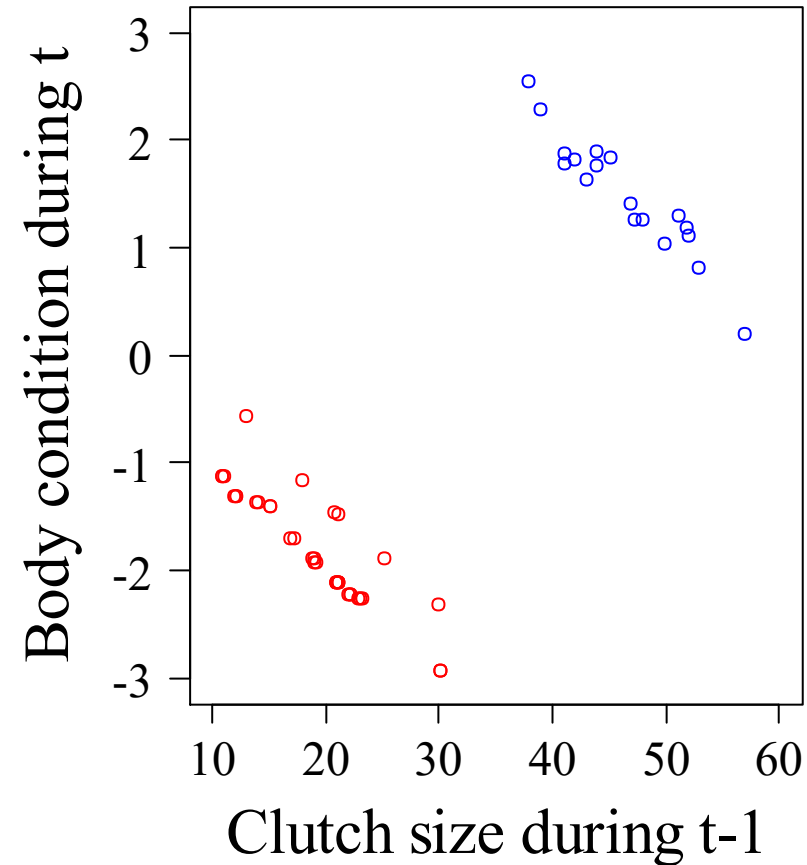
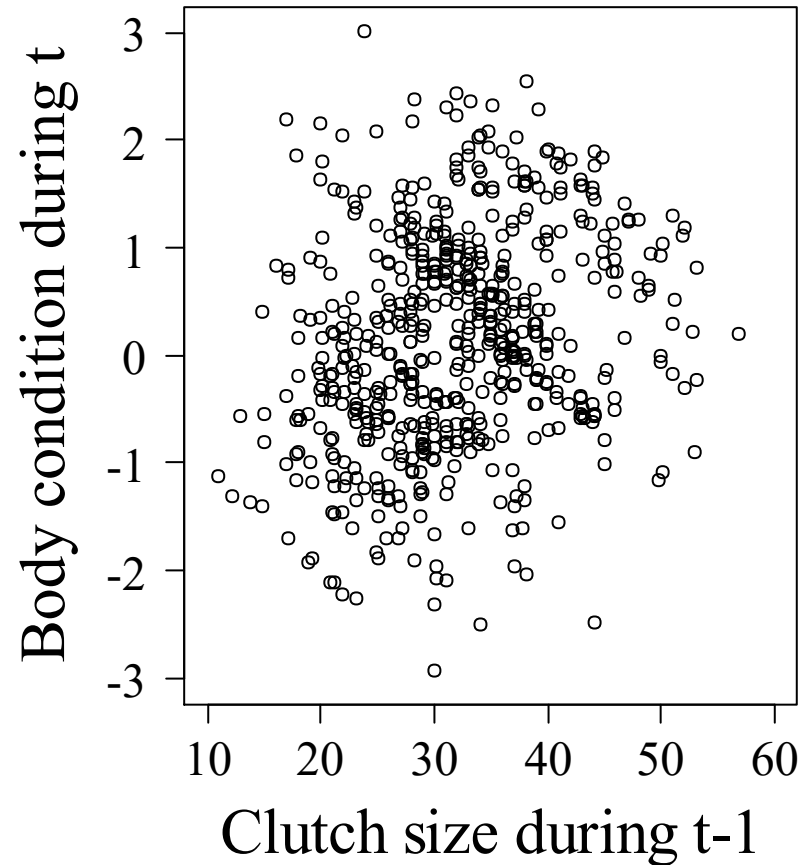
Here are the two 'worst' gators

What's happening here?!



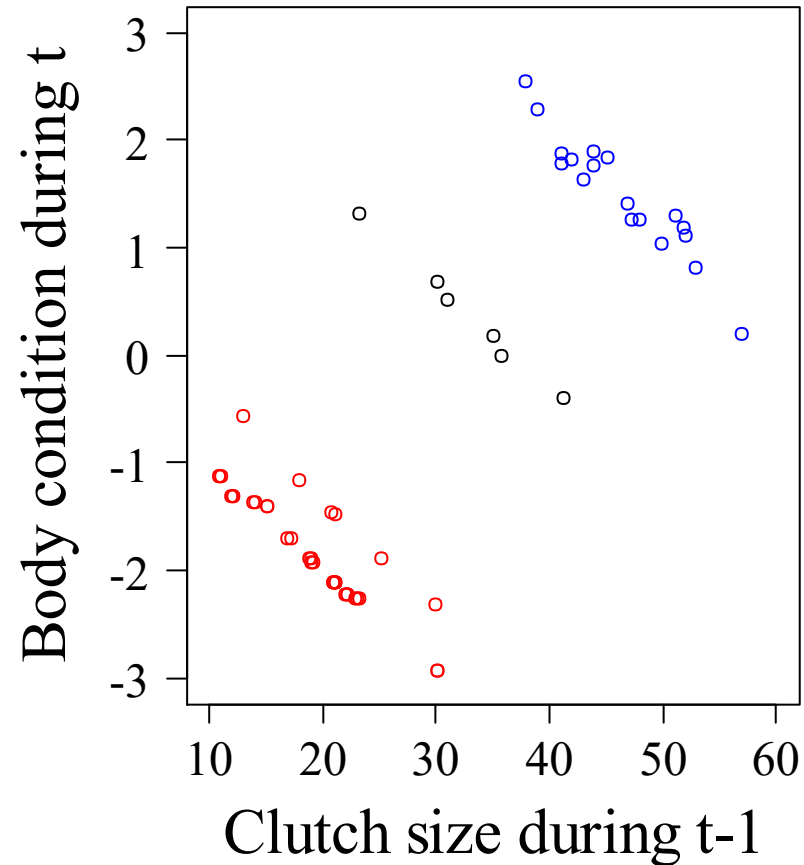
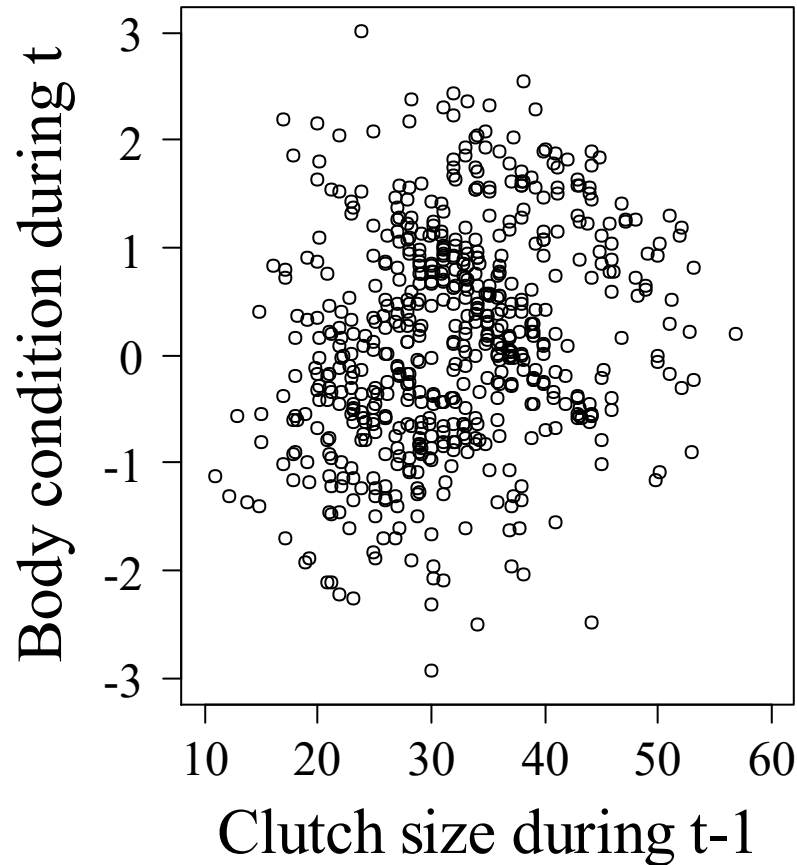
Here are the three 'worst' gators

What's happening here?!



Here are the three 'worst' and three 'best' gators

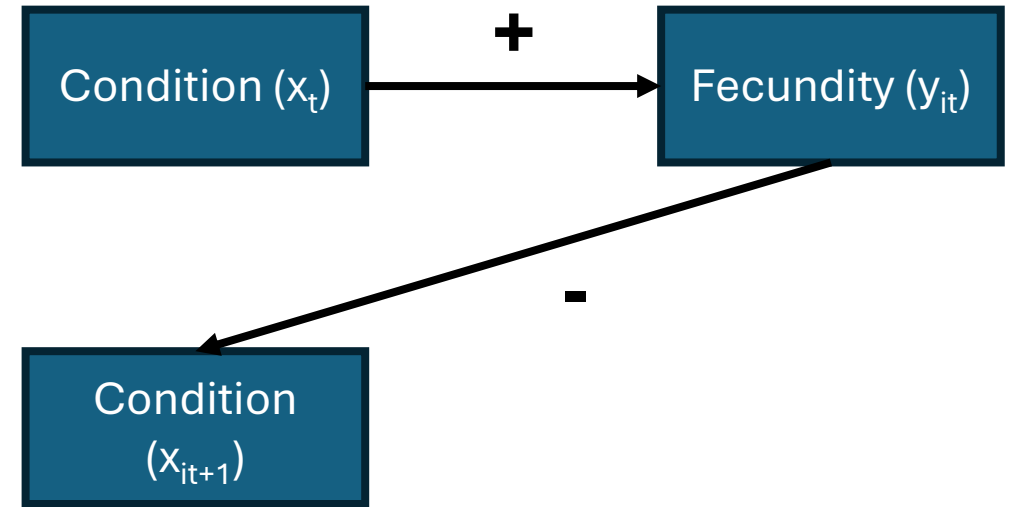
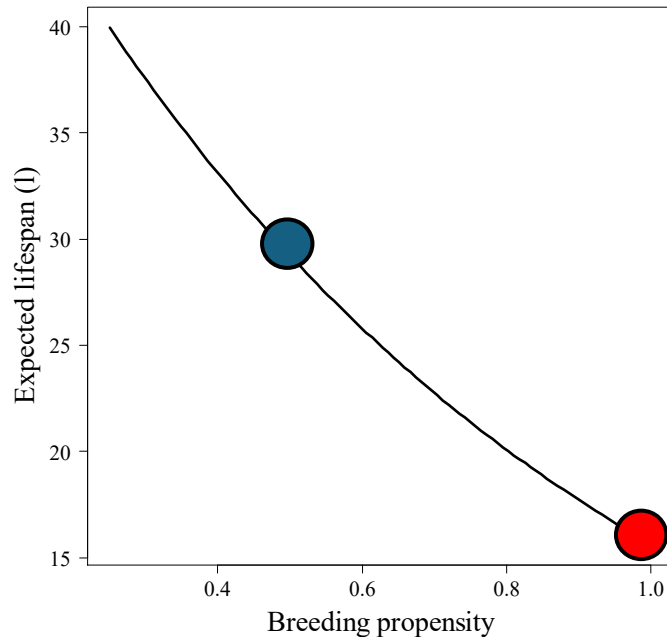
What's happening here?!



Here are the three 'worst', three 'best', and median gators

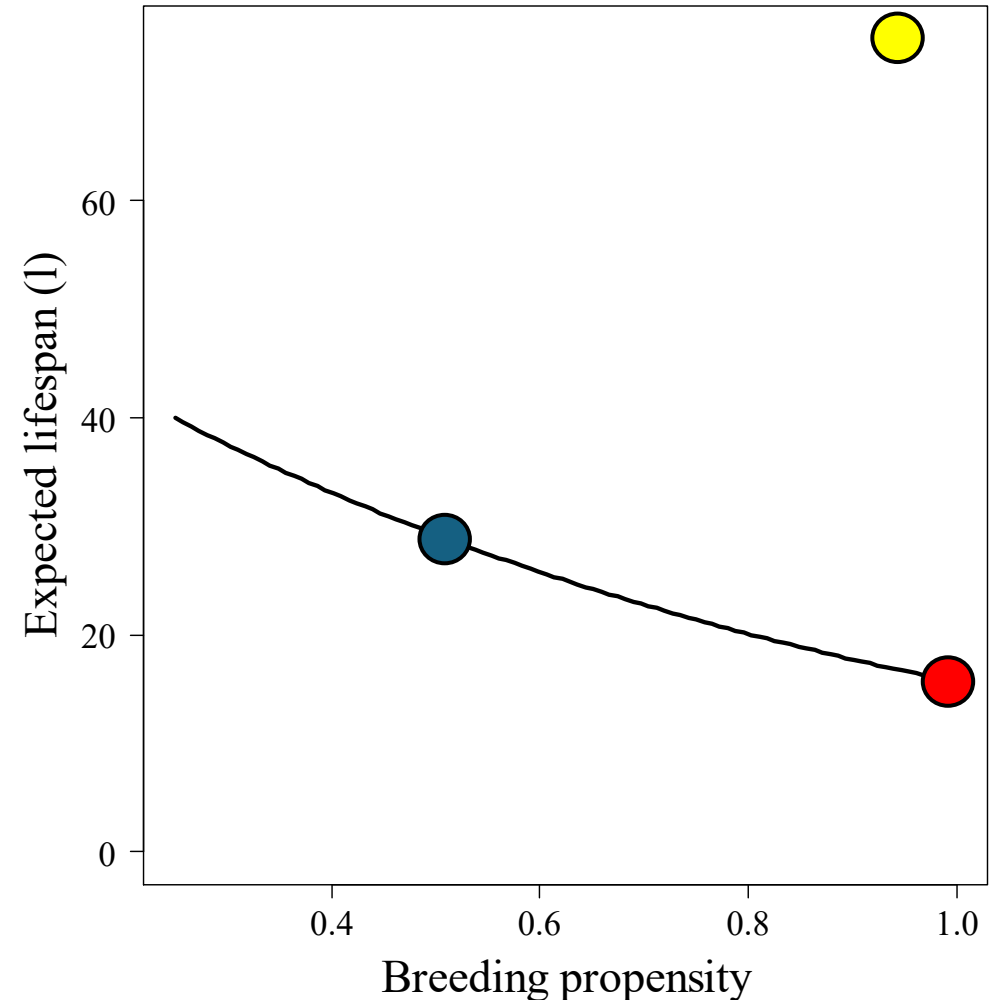
A real-life example: Laysan albatross

- Long-lived (30+ years)
- Delayed maturity (7-9 years)
- Lay a single egg
- Incubation (2 months)
- Fledging (5 months)

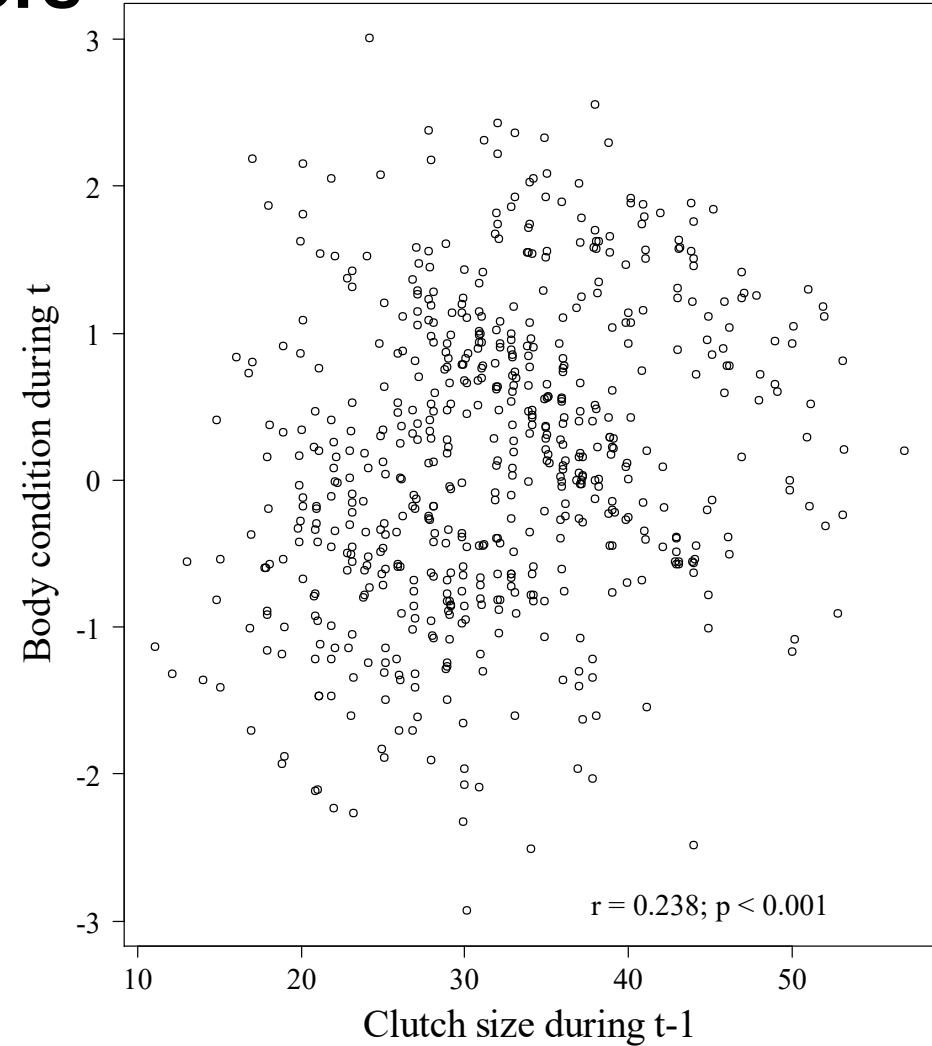
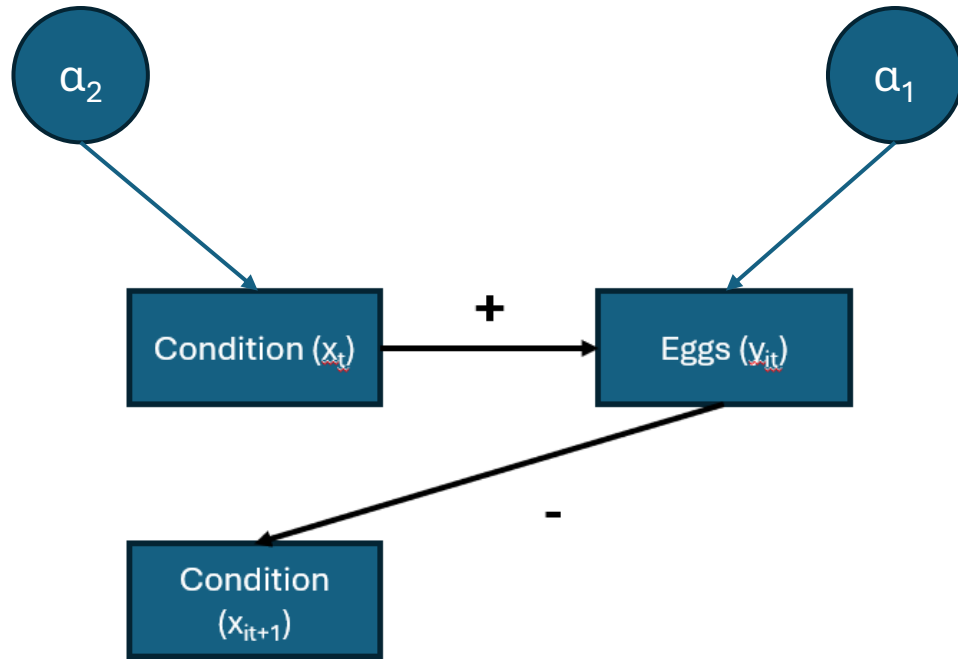


A real-life example: 'Wisdom' the Laysan albatross

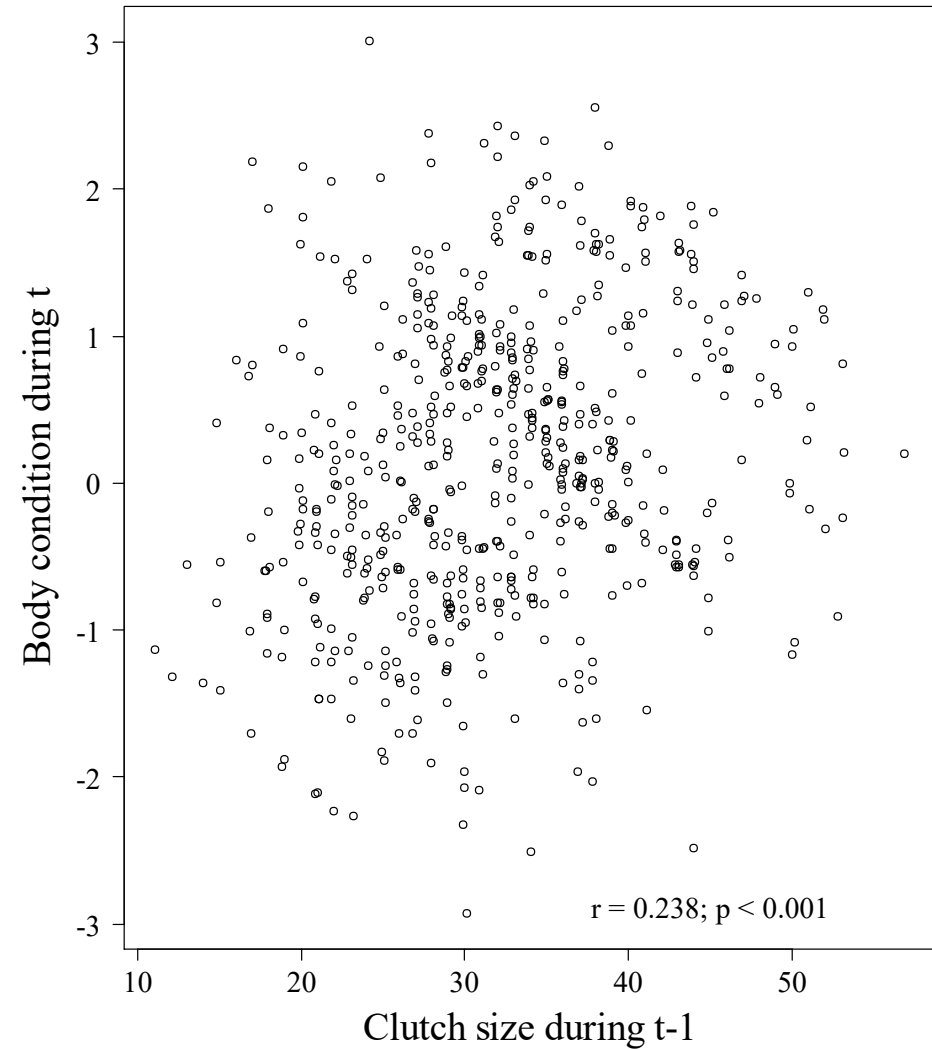
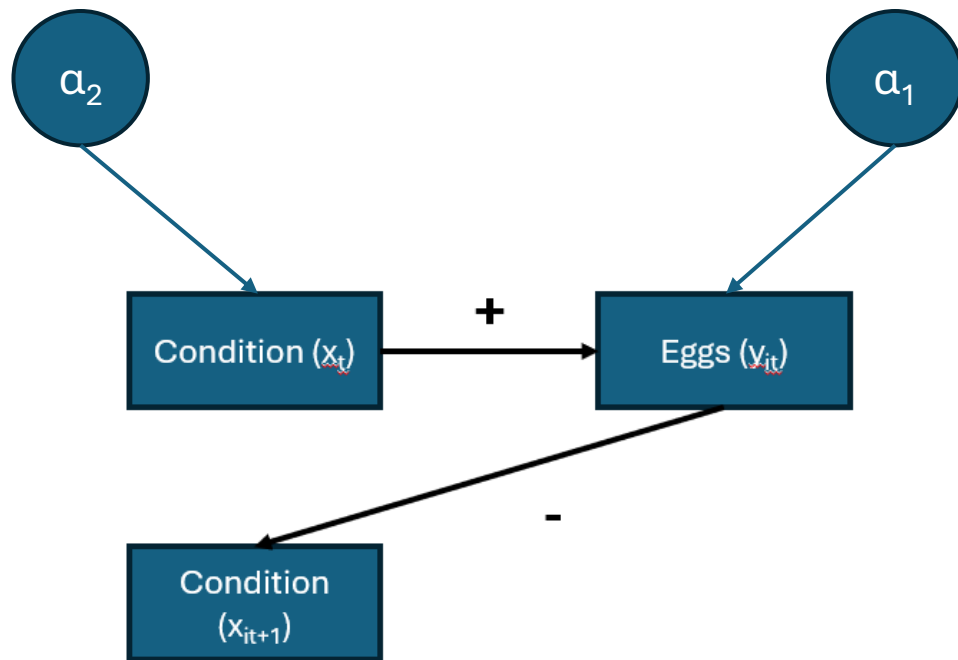
- First marked as an adult in 1956 (74+ years old)
- Oldest known living wild bird
- Breeds almost every year (50-60 eggs, 30+ fledglings)



Let's come back to simulated alligators



We need to account for heterogeneity



Seeing how analyses break helps us understand how they work

Take-homes

1. Cross-lags can lead to 'wrong answers' (e.g., alligator simulation, pintails) that defy theory.

Take-homes

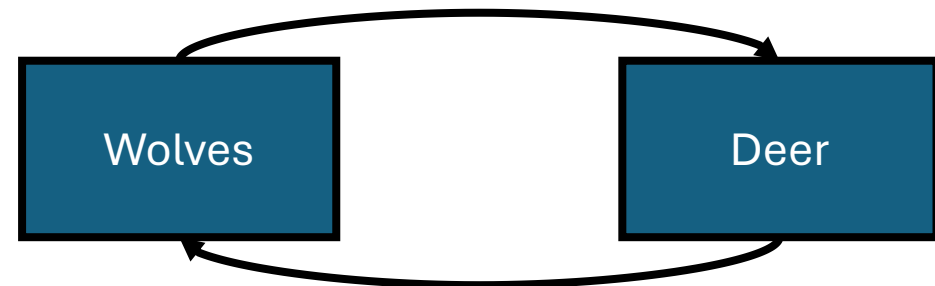
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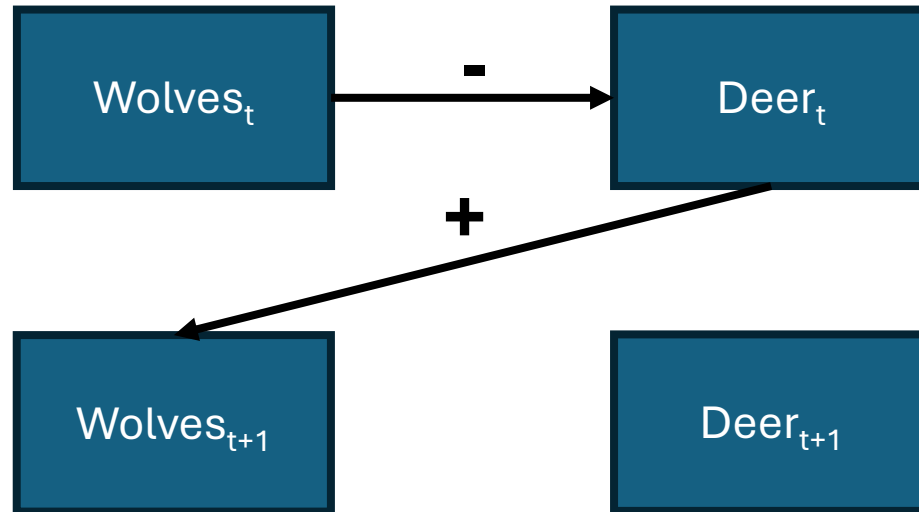


Take-homes

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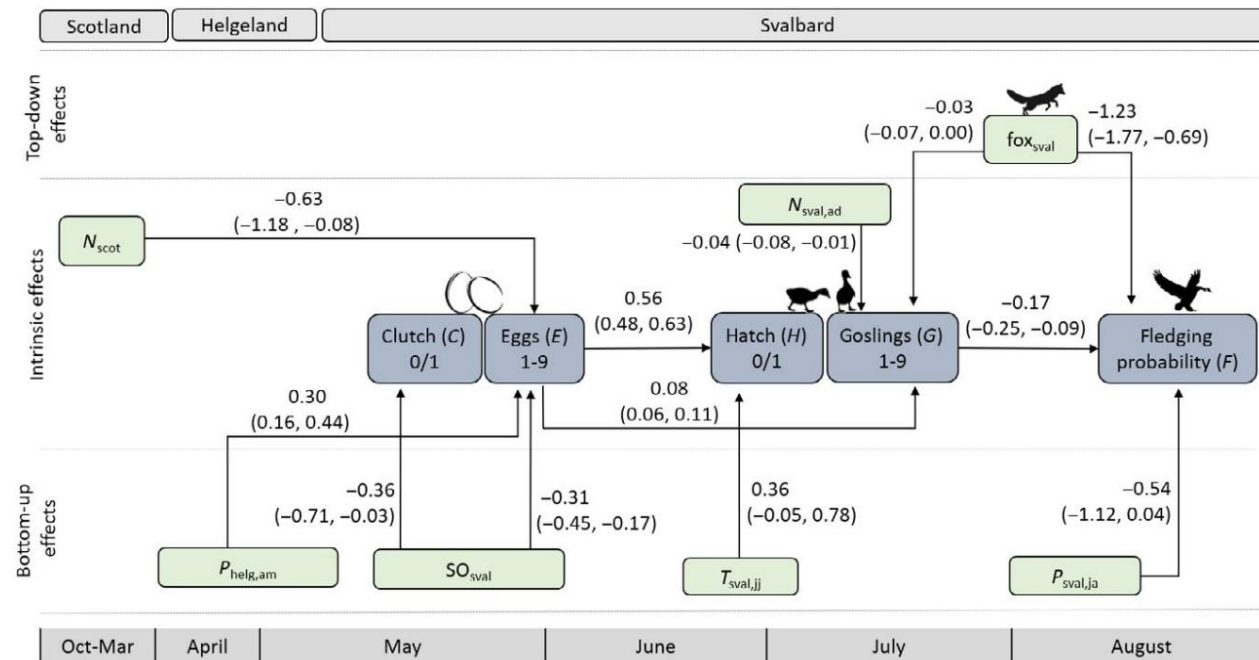
$$\frac{\delta d}{\delta t} = f(w)$$

$$\frac{\delta w}{\delta t} = f(d)$$



Take-homes

4. We don't have to use 'cross-lags' across years, simply modeling sequential processes within a season can be extremely valuable



Layton-Matthews et al. (2019) *Global Change Biology*